ETC Installation Guide
ColorSource Linear v1.7

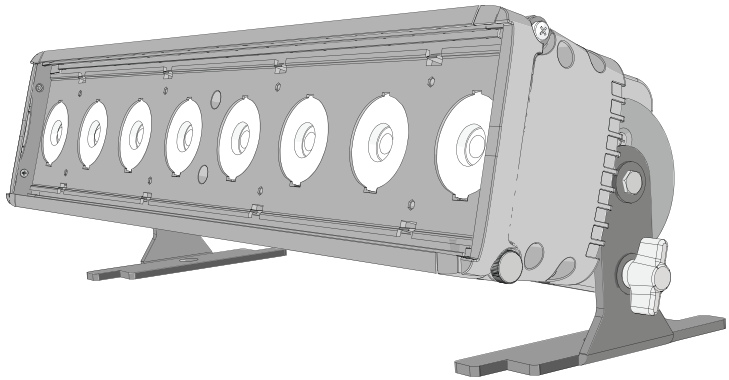
Overview

The ColorSource Linear is an affordable LED fixture that uses the RGB-L (red, green, blue, and lime) color system to provide a rich, bright light.

The fixture is available in three lengths:

- ColorSource Linear 1 (0.5 meter)
- ColorSource Linear 2 (1 meter)
- ColorSource Linear 4 (2 meter)

The ColorSource Linear is available in both original and Deep Blue variants. The Deep Blue uses an array that offers richer, more saturated blues and stronger magentas in exchange for the more jewel-toned blues like sky or turquoise. For more information, see the ETC website at etcconnect.com.



Note: *Features described in this document require the latest version of software. For information on updating fixture software, see* *Update the Fixture Software on page 15*.

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Safety

For professional use only. Read the entire manual before using this equipment.

WARNING:

- **Risk of eye injury. Do not stare directly into the light.**
- **Do not mount the fixture on or near a flammable surface.**
- **Disconnect the fixture from power and DMX and allow it to cool before installing accessories or performing any cleaning and maintenance.**

AVERTISSEMENT :

- **Risque de lésion oculaire. Ne fixez pas directement la lumière.**
- **Ne pas installer le projecteur sur ou à côté d'une surface inflammable.**
- **Débranchez le projecteur de son alimentation et du DMX et laissez-le refroidir avant d'installer des accessoires ou d'effectuer un nettoyage ou un entretien.**

Note: *The light source in this luminaire must only be replaced by the manufacturer, the manufacturer's service agent, or a similarly qualified person.*

Note:

- *DMX512 connections must be Class 2.*
- *To ensure that cumulative leakage current on control wiring does not exceed 3.5 mA, connect no more than 32 fixtures on a single DMX run.*

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The following symbols may appear on product labeling.

	The luminaire must be installed at least 0.1 m (4.0 in) away from all lighted objects.	Le luminaire doit être installé à au moins 0,1 m (4,0 po) de tout objet éclairé.
	Do not mount the fixture on or near a flammable surface.	Ne pas installer le projecteur sur ou à côté d'une surface inflammable.
	Do not stare at the operating light source.	Ne pas regarder la source de lumière lorsqu'elle fonctionne.
	This product should not be discarded as unsorted waste but must be sent to separate collection facilities for recovery and recycling.	Ce produit ne doit pas être jeté avec les déchets ménagers mais doit être déposé dans une collecte de déchets électroniques ou dans un point de collecte.
	Protective earth (ground)	Protection Classe I Mise à la terre

Installation

Mounting

The ColorSource Linear ships with a pair of floor trunnions for mounting.

Note: *Ensure that the trunnions are installed with the feet inward so that fixtures can be lined up end-to-end, if needed.*

You also have the option of using clamps with the trunnions to hang the fixture. Attach the clamps to the holes in the trunnion feet (see figure on *page 4*).

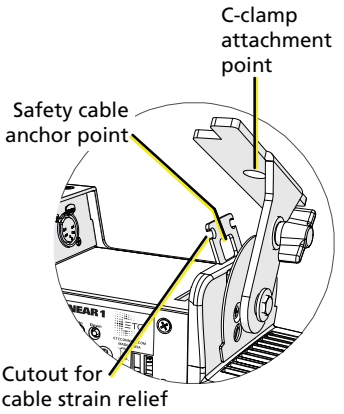
The ColorSource Linear 1 has two additional mounting options. Contact your local dealer for more information.

- Yoke with C-clamp (7414K1017)
- Double yoke with C-clamp for mounting two fixtures (7414K1018)

Safety Cable

Two safety cable anchor points are provided on the fixture housing.

The anchor points also include two cutouts that can be used for cable strain relief.



Connectors

- **Power:** Two powerCON® connectors for power in and thru
- **Data (DMX512/RDM):** Two five-pin XLR connectors OR two RJ45 connectors for data in and thru

Conversion cables are available from ETC. Please contact your ETC customer service representative and request part number W6538 for RJ45 to socket XLR or part number W6539 for RJ45 to plug XLR.

Connector locations

- **Linear 1:** Power connectors on one side of the fixture and data connectors on the other
- **Linear 2 and Linear 4:** Power and data connectors on each end (to facilitate daisy-chaining)

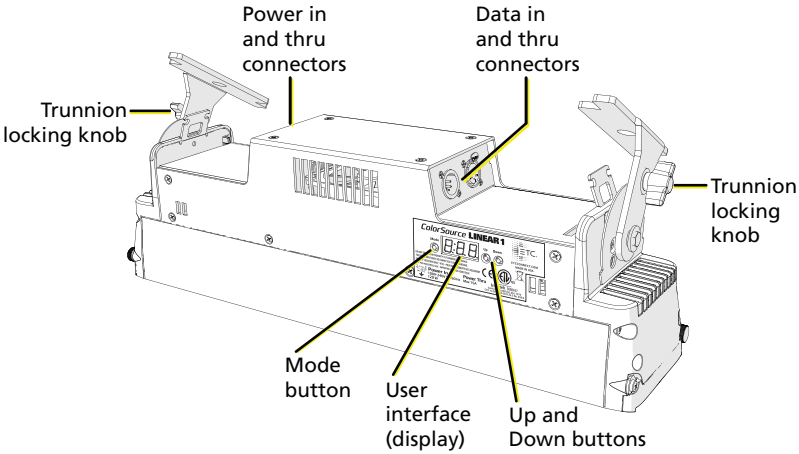
Note: *This fixture is not Ethernet/data network compatible.*

Note: *The ColorSource Linear is not self terminating. You must terminate the last fixture in line with a 120 Ohm resistor. To purchase a terminator, please contact ETC customer service and request part number N4086 (RJ45 terminator) or SGE1507 (XLR terminator).*

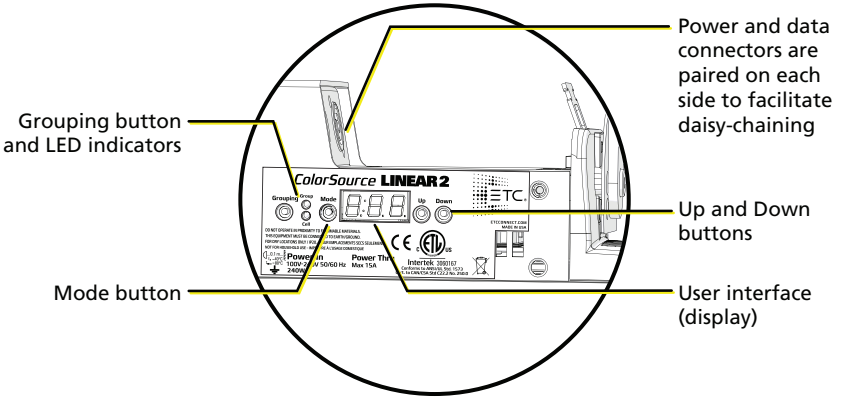
Set Up the Fixture

See the fixture illustrations below and on the following page for reference.

1. Plug the power cord into the powerCON In connector on the rear of the fixture.
2. Make any power-thru cable connections prior to applying a power source.
3. Make any DMX connections, and if the fixture is the last in a DMX line, terminate the fixture. See *Connectors on page 4*.
4. Plug the power cord into the power source.



Linear 1



Linear 2 and Linear 4

Power Up

Press any button to wake the display. The display shows the DMX address. See *Set the DMX Address on page 7*.

You can press the **Mode** button to access the Presets/Sequences menu. See *Play Presets and Sequences on page 13*.

Adjust the Tilt

The fixture can be tilted up and down as needed.

1. Loosen, but do not remove the two trunnion locking knobs.
2. Tilt the fixture to the desired angle.
3. Tighten the trunnion locking knobs.

Set the DMX Address

1. Press any button to wake the display. The display shows the DMX address.
2. Use the **Up** and **Down** buttons to select the DMX address for the fixture.

You can also use RDM for DMX addressing. See *Use RDM*.

Change the DMX Control Mode

You can set the DMX control mode for the fixture to one of four available mode options.

1. Press any button to wake the display. The display shows the DMX address.
2. Press and hold the **Mode** button for 3 seconds. The display shows the current DMX control mode.
3. Press the **Up** button or **Down** button to change the DMX control mode. (The currently saved mode is indicated by a decimal point at the end of the mode name.)
   - **5ch**: 5-channel mode (intensity, red, green, blue, and strobe). This is the default mode.
   - **d ir**: Direct emitter control (intensity, red, green, blue/indigo, lime, and strobe).
   - **1ch**: 1-channel mode controls the intensity of preset 1 (defaults to 3200 K White).
   - **rgb**: RGB mode (red, green, and blue).
4. Press the **Mode** button to confirm your selection.

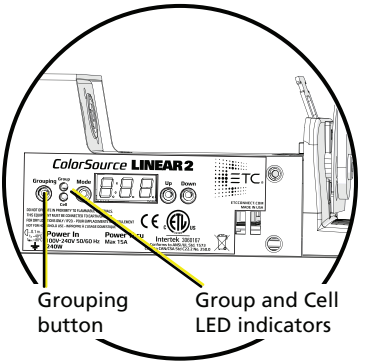
   If you do not press the **Mode** button, the display will automatically return to the DMX address after 10 seconds of inactivity, and edits will not be saved. This cancels unsaved changes.

Set the Grouping Mode

The ColorSource Linear 2 and the ColorSource Linear 4 fixtures have two Grouping modes:

- **Group:** Entire array functions as a single fixture.
- **Cell:** Each 0.5 m segment (or "cell") of the array functions as a distinct fixture with a separate DMX start address.

To set the Grouping mode, press the **Grouping** button until the LED indicator for the chosen mode illuminates.



In Cell mode, each cell is assigned a DMX address based on the DMX address of the fixture (see *Set the DMX Address on page 7*) and the current DMX control mode (see *Change the DMX Control Mode on page 7*).

For example, if you set the DMX address on the fixture to 100, set the DMX mode to 5-channel, and set the Grouping mode to Cell, the starting DMX address of the first cell will be 100 and the starting DMX address of the second cell will be 105.

DMX Control Channels

Linear 1 and Linear 2/Linear 4 in Group Mode

DMX Channel	5ch (5 channel)	dir (Direct)	1ch (1 Channel)	rgb (RGB)
1	Intensity	Intensity	Intensity	Red
2	Red	Red		Green
3	Green	Green		Blue
4	Blue	Blue/Indigo*		
5	Strobe**	Lime		
6		Strobe**		

*Original ColorSource Linear fixture uses blue in channel 4; Deep Blue fixture uses indigo in channel 4.

**Use DMX values of 0 or 255 for no strobe. Use values of 1–254 to increase the strobe rate from 1 per second up to 33 per second.

Linear 2 in Cell Mode

DMX Channel	5ch (5 Channel		d ir (Direct)		Ich (1 Channel)		r9b (RGB)	
1	Cell 1	Intensity	Cell 1	Intensity	Cell 1	Intensity	Cell 1	Red
2		Red		Red	Cell 2	Intensity		Green
3		Green		Green				Blue
4		Blue		Blue/Indigo*			Cell 2	Red
5		Strobe**		Lime				Green
6	Cell 2	Intensity		Strobe**				Blue
7		Red	Cell 2	Intensity				
8		Green		Red				
9		Blue		Green				
10		Strobe**		Blue/Indigo*				
11				Lime				
12				Strobe**				

*Original ColorSource Linear fixture uses blue in channel 4; Deep Blue fixture uses indigo in channel 4.

**Use DMX values of 0 or 255 for no strobe. Use values of 1–254 to increase the strobe rate from 1 per second up to 33 per second.

Linear 4 in Cell Mode

DMX Channel	5ch (5 Channel		d ir (Direct)		Ich (1 Channel)		rgb (RGB)	
1	Cell 1	Intensity	Cell 1	Intensity	Cell 1	Intensity	Cell 1	Red
2		Red		Red	Cell 2	Intensity		Green
3		Green		Green	Cell 3	Intensity		Blue
4		Blue		Blue/Indigo*	Cell 4	Intensity	Cell 2	Red
5		Strobe**		Lime				Green
6	Cell 2	Intensity		Strobe**				Blue
7		Red	Cell 2	Intensity			Cell 3	Red
8		Green		Red				Green
9		Blue		Green				Blue
10		Strobe**		Blue/Indigo*			Cell 4	Red
11	Cell 3	Intensity		Lime				Green
12		Red		Strobe**				Blue
13		Green	Cell 3	Intensity				
14		Blue		Red				
15		Strobe**		Green				
16	Cell 4	Intensity		Blue/Indigo*				
17		Red		Lime				
18		Green		Strobe**				
19		Blue	Cell 4	Intensity				
20		Strobe**		Red				
21				Green				
22				Blue/Indigo*				
23				Lime				
24				Strobe**				

*Original ColorSource Linear fixture uses blue in channel 4; Deep Blue fixture uses indigo in channel 4.

**Use DMX values of 0 or 255 for no strobe. Use values of 1–254 to increase the strobe rate from 1 per second up to 33 per second.

Data Loss Behavior

When DMX data is lost, the fixture does one of the following actions:

- If you selected a preset or sequence, that preset or sequence automatically plays.
- If no preset or sequence is active in the background, the fixture goes dark and the DMX start address flashes on the user interface.

Lock the User Interface

After you have set up the fixture, you can lock the user interface to prevent additional changes.

1. Press and hold the **Up** button and **Down** button simultaneously for 3 seconds. The display will show **Loc** to indicate that the user interface is locked.
2. To unlock the user interface, press and hold the **Up** button and **Down** button simultaneously for 3 seconds.

Accessories

You can use several half-meter optional accessories with ColorSource Linear fixtures, including:

- Secondary lenses (horizontal, vertical, and round)
- Barn doors (7414K1020)
- Louver (7414K1019)

For information on installing secondary lenses, see *Install Secondary Lenses on page 12*.

Accessories for Linear 2 and Linear 4

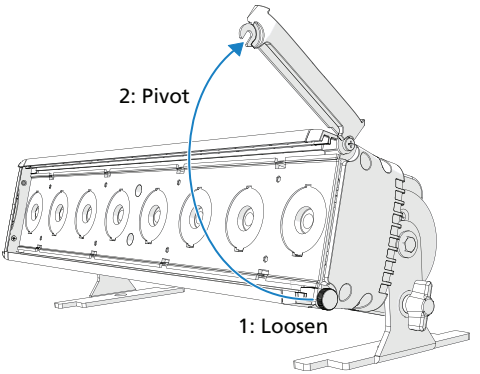
For the Linear 2 and Linear 4 fixtures, order one accessory for each 0.5 meter segment (cell) of the fixture. For example, for a Linear 2 fixture, order two barn door accessory kits to fit the full length of the fixture.

Spacers for secondary lenses are included with the fixture. Use these spacers to hold the accessories in position when multiple lenses are used in the Linear 2 and Linear 4 fixtures. The spacers are positioned between the 0.5 meter lenses.

Contact your local dealer for more information.

Install Secondary Lenses

You can use several optional accessories with the ColorSource Linear, including secondary lenses (horizontal, vertical, and round). You can use either a single lens or two lenses simultaneously, and you can install them from either end of the fixture.

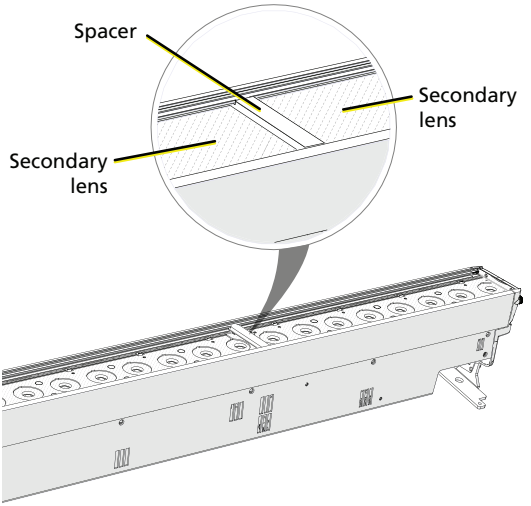


For the Linear 2 and Linear 4 fixtures, order one accessory for each 0.5 meter segment (cell) of the fixture.

Spacers for secondary lenses are included with the fixture. Use these spacers to hold the accessories in position when multiple lenses are used in the Linear 2 and Linear 4 fixtures.


Spacer

1. Loosen the thumbscrew on the access panel on either end of the fixture.
2. Pivot the access panel until you can access the slots above the LEDs.



3. Slide the secondary lens (or lenses) into either of the slots above the LEDs.
4. For the Linear 2 and Linear 4 fixtures, position spacers between the lenses for the cells (in the same slot as the lenses). See image at left.
5. Pivot the access panel back into place and tighten the thumbscrew.

Play Presets and Sequences

The fixture has twelve factory-configured presets and five factory-configured sequences. When you play a preset or sequence on a fixture:

- The same preset number or sequence number plays on any connected fixtures that are not receiving DMX.
- If you power down the fixture, the preset or sequence resumes when you power it on again.

You can re-record presets and sequences through an RDM device or manually on the fixture. See *Re-record Presets on page 14*.

When a Linear 2 or Linear 4 fixture is in Cell mode, you can re-record a preset with each cell having a distinct set of values. See *Set the Grouping Mode on page 8*.

1. Press the **Mode** button to activate the Presets/Sequences menu. The last active preset or sequence plays automatically. (If DMX data is present, the current preset or sequence plays in the background and DMX data continues to take priority.)
2. Press the **Up** button or **Down** button as needed to scroll through the available presets and sequences.
3. To stop the selected preset or sequence, press the **Mode** button a second time. The display will revert to show the DMX start address.

Pre-configured Presets

Preset	Name
1	3200 K White
2	5600 K White
3	Red
4	Green
5	Blue
6	Cyan

Preset	Name
7	Magenta
8	Yellow
9	Orange
10	Light Yellow
11	Light Blue
12	Light Pink

Pre-configured Sequences

Sequence	Name
1	Rainbow Fast
2	Rainbow Slow
3	Warm Colors Fast
4	Cool Colors Fast
5	Random

Re-record Presets

1. Using DMX, set the desired levels for the new preset.
2. Press the **Mode** button to access the Presets/Sequences menu.
3. Press the **Up** button or **Down** button to navigate to the preset number that you want to re-record.
4. Press and hold the **Mode** button for 3 seconds to confirm the preset selection. The user interface will show a 3-second countdown. The preset number blinks once to confirm that the preset has been recorded.

Note: *When the fixture is in Direct mode (d ir) or 1-channel mode (1ch), you cannot re-record presets.*

When you re-record presets with the fixture in RGB mode (rGb), the intensity is set to full and the strobe is turned off.

See Change the DMX Control Mode on page 7.

Adjust the Intensity of Presets

1. Ensure that DMX data is not present.
2. Press the **Mode** button to access the Presets/Sequences menu, and then press the **Up** button or **Down** button to select the preset that you want to edit.
3. Press and hold the **Mode** button for 3 seconds. The display shows L: followed by either an intensity level from 0- 99 or FL for full.
4. Press the **Up** button or **Down** button to change the intensity level of the preset in real time.
5. Press the **Mode** button to confirm your selection and return to the Presets/Sequences menu. (If you don't press the **Mode** button, the display returns to the Presets menu after 10 seconds without saving your changes.)

Troubleshooting

Update the Fixture Software

ETC recommends using UpdaterAtor software to manage software updates. For information on UpdaterAtor, see the *UpdaterAtor Software Quick Guide* and the *UpdaterAtor Software Release Note*, which you can download from etcconnect.com.

You can push newer software versions from a connected fixture to selected ColorSource fixtures (see table that follows).

Note: *If you have other types of fixtures in your DMX run, you do not need to disconnect them before pushing the updated fixture software. The update process will ignore any other fixture types in the DMX run.*

Note: *Not sure what version of fixture software is installed? The version number appears on the display when the fixture powers up.*

This fixture...	...can push software to these fixtures/releases when connected								
	CYC			Linear	PAR			Spot	
	≤v1.7	v3.1	v4.0	≤v1.7	≤v1.7	v3.0	v4.0	≤v1.7	v3.0
CYC v1.7 or earlier	X			X	X			X	
CYC v3.1		X							
CYC v4.0			X						
Linear v1.7 or earlier	X			X	X			X	
PAR v1.7 or earlier	X			X	X			X	
PAR v3.0						X			X
PAR v4.0							X		
Spot v1.7 or earlier	X			X	X			X	
Spot v3.0						X			X

1. Remove power and DMX In from the fixture that is running the software version that you want to push to other fixtures.
2. Press and hold the **Up** button and **Down** button simultaneously while restoring power.
3. The display shows **Udt rdy** to indicate the update mode is selected.

4. Press the **Mode** button to start the update process.
    - The display alternates between **UdE** and the percentage complete while updating the connected fixtures.
    - The display returns to its previous value (DMX address or active preset) to indicate that the update is complete.
5. Reconnect DMX In on the fixture.

Restore Factory Defaults

Press and hold the **Mode** button, **Up** button, and **Down** button simultaneously for 3 seconds. The DMX start address will blink once to confirm that defaults have been restored.

Error Codes

These error codes may display on the user interface.

- **CAL**: Screen toggles between **CAL** and **Err** to indicate a problem with the color calibration. The fixture will need to be returned to the ETC factory; contact your local dealer to set up a repair.
- **OtP**: Screen toggles between **OtP** and **Err** to indicate that the fixture has gone into over-temperature protection mode to avoid damaging the fixture. Once the fixture has self-cooled, the error will automatically clear. If the error does not clear, contact ETC Technical Support (visit etcconnect.com/contactETC).
- **CbL**: Screen toggles between **CbL** and **Err** to indicate that a control card or the internal cabling is damaged or unplugged. Verify that internal cables are properly connected. If the error persists, the fixture will need to be returned to the ETC factory; contact your local dealer to set up a repair.
- **UdE Err**: Screen toggles between **UdE** and **Err** to indicate that the software update failed. Reload software to clear the error. If the error persists, contact ETC Technical Support (visit etcconnect.com/contactETC).
- **no APP**: Screen toggles between **no** and **APP** to indicate that the upgraded fixtures have a software issue. Reload software to clear the error. If the error persists, contact ETC Technical Support (visit etcconnect.com/contactETC).

Specifications and Reference

For current and complete compliance and specifications, see the ColorSource Linear datasheet at etcconnect.com.

Environment

The ColorSource Linear operates in ambient temperatures of 0°C–40°C and comes with a variable-speed fan. The ColorSource Linear is for indoor use only.

- Maximum recommended ambient operating temperature: Ta = 40°C (104°F)
- Maximum anticipated external surface temperature at 40°C ambient: Tmax = 80°C (176°F)
- External temperatures after 5 minutes of full-brightness operation at 25°C (77°F) ambient: 36°C (97°F)
- External Temperature (steady state achieved) at 25°C (77°F): 48°C (118°F)

Electrical

- Operates between 100 V and 240 VAC at a frequency of 50/60 Hz.
- Maximum power consumption:
  - Linear 1: 120 W
  - Linear 2: 240 W
  - Linear 4: 480 W
- Up to 9 cells (0.5 m segments) may be linked via power thru connector.
- Up to 10 cells (0.5 m segments) total per circuit when used with an R20 Relay Module or Unison Echo Relay Panel. Consult breaker trip curves when used with other equipment. Requires power from a non-dimmable source.
- Inrush (Linear 1/Linear 2/Linear 4):
  - 120 V: 35 A/35.6 A/67.5 A (first half-cycle)
  - 240 V: 49 A/85.6 A/153 A (first half-cycle)

Typical Power Consumption

Linear 1	100 V		120 V		230 V	
	Power	Current	Power	Current	Power	Current
Idle	1.63 W	0.03 A	1.55 W	0.03 A	1.43 W	0.03 A
Direct at Full	107 W	1.07 A	105 W	0.88 A	104 W	0.46 A

Linear 2	100 V		120 V		230 V	
	Power	Current	Power	Current	Power	Current
Idle	2.86 W	0.06 A	2.98 W	0.06 A	2.56 W	0.06 A
Direct at Full	210.83 W	2.12 A	208.67 W	1.75 A	204.5 W	0.91 A

Linear 4	100 V		120 V		230 V	
	Power	Current	Power	Current	Power	Current
Idle	5.28 W	0.11 A	5.24 W	0.11 A	4.65 W	0.12 A
Direct at Full	424.17 W	4.27 A	419.33 W	3.52 A	409.17 W	1.82 A

Use RDM

RDM and Grouping Modes

Linear 2 and Linear 4 fixtures in Cell mode contain 2 sub-devices (Linear 2) or 4 sub-devices (Linear 4). Use DMX_BLOCK_ADDRESS to set the DMX address of the first cell, and the remaining cells will be numbered sequentially from that starting value.

RDM also allows you to use nonsequential DMX addresses, if desired. The fixture displays "---" for the DMX address to indicate nonsequential DMX addressing. If you set up nonsequential DMX addresses and then change the DMX address at the fixture, however, the fixture will return to sequential DMX addressing.

Linear 2 and Linear 4 fixtures in Group mode do not use sub-devices.

RDM Values

Parameter	Fixture	Value	Description
Manufacturer ID	All	0x6574	Electronic Theatre Controls
Model ID	Linear 1	0x0209	
	Linear 1 Deep Blue	0x020A	
	Linear 2	0x020D	
	Linear 2 Deep Blue	0x020E	
	Linear 4	0x0211	
	Linear 4 Deep Blue	0x0212	
DMX Start Address	All	0x00F0	Range = 1–512
DMX Personality	Linear 1	0x00E0	1 = 5 Channel 2 = Direct 3 = 1 Channel 4 = RGB
	Linear 2 in Group mode		
	Linear 4 in Group mode		
	Linear 2 in Cell mode		5 = 5 Channel Multi-Cell 6 = Direct Multi-Cell 7 = 1 Channel Multi-Cell 8 = RGB Multi-Cell
	Linear 4 in Cell mode		
Output Frequency	Linear 1	0x8123	0 = Standard (1.2 kHz) 1 = High (25 kHz)
	Linear 2 in Group mode		
	Linear 4 in Group mode		

FCC Compliance

ColorSource Linear

(For any FCC matters):

Electronic Theatre Controls, Inc.
3031 Pleasant View Road
Middleton, WI 53562
+1 (608) 831-4116
etcconnect.com

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) this device may not cause harmful interference, and (2) this device must accept any interference received; including interference that may cause undesired operation. Visit etcconnect.com/products for current and complete compliance information including FCC compliance.



Note: *This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to Part 15 of the FCC rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. Any modifications or changes to this product not expressly approved by Electronic Theatre Controls, Inc. could void the user's authority to operate the product. Operation of this equipment in a residential area is likely to cause harmful interference, in which case the user will be required to correct the interference at their own expense.*